



unEF009

S.V. National Institute of Technology, Surat
Department of Electronics Engineering
MID Semester Exam, SEPTEMBER 2024
B. Tech. (EC) 5th Semester
Digital Communication (EC 305)

Marks: 30

Time: 1.5 Hours

Instructions:

1. Answer all the questions in the sequential order
2. Assume data wherever required

ATTEMPT THE FOLLOWING QUESTIONS

Marks

- | 1. | A discrete memoryless source is described by the alphabet {A, B, C, D, E} with probabilities {1/32, 1/8, 1/2, 1/16, 1/32, 1/4} respectively.
a. Design a Huffman code for the above source and find the average length of the code word
b. Can you improve the Huffman code designed in part a. by encoding the second order extension of the source? Give reason(s) for your answer. | 3 | CO1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|----|---|-----|------|-----|-----|---|-----|---|-----|--|---|--|--|--|--|--------|---|---|---|---|---|-----|-----|-----|---|-----|-----|-----|---|-----|-----|---|---|-----|
| 2. | The parity check equations for a (6,3) systematic code are
$b_0 = c_3 = m_0 + m_1 + m_2$ $b_1 = c_4 = m_0 + m_1$ $b_3 = c_5 = m_0 + m_2$
a. Determine the generator matrix of the code
b. Determine the parity check matrix of the code
c. List out all the code vectors
d. What is the error correcting capability of the code?
e. Decode the received codeword 101100 | 5 | CO1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3. | Taking $x^3 + x + 1$ as a generator polynomial for the (7,4) cyclic linear block, determine the code vectors in systematic form for the following message sequences
a. 1 0 1 1
b. 1 0 0 0 | 2 | CO3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4. | A source produces three symbols A, B and C with the following marginal and conditional probabilities
<table border="1" style="display: inline-table; margin: 10px;"><thead><tr><th>i</th><th>p(i)</th></tr></thead><tbody><tr><td>1</td><td>1/4</td></tr><tr><td>2</td><td>1/4</td></tr><tr><td>3</td><td>1/2</td></tr></tbody></table> <table border="1" style="display: inline-table; margin: 10px;"><thead><tr><th></th><th colspan="4">j</th></tr><tr><th></th><th>p(j i)</th><th>A</th><th>B</th><th>C</th></tr></thead><tbody><tr><th rowspan="3">i</th><th>A</th><td>1/8</td><td>1/4</td><td>5/8</td></tr><tr><th>B</th><td>1/2</td><td>1/8</td><td>3/8</td></tr><tr><th>C</th><td>3/8</td><td>5/8</td><td>0</td></tr></tbody></table>
a. Assuming that there is no inter-symbol influence, calculate the entropy of the source
b. If index 'i' refers to the X and index 'j' refers to Y, determine the conditional entropy H(Y X) | i | p(i) | 1 | 1/4 | 2 | 1/4 | 3 | 1/2 | | j | | | | | p(j i) | A | B | C | i | A | 1/8 | 1/4 | 5/8 | B | 1/2 | 1/8 | 3/8 | C | 3/8 | 5/8 | 0 | 3 | CO3 |
| i | p(i) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1 | 1/4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2 | 1/4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3 | 1/2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | j | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | p(j i) | A | B | C | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| i | A | 1/8 | 1/4 | 5/8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | B | 1/2 | 1/8 | 3/8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | C | 3/8 | 5/8 | 0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5. | Find the channel capacity of a binary symmetric channel. Sketch C_s vs. p , the transition probability given that input symbols are equiprobable. | 4 | CO1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6. | Using 4×10^5 pixels per frame, with each pixel taking any one of the 10 different brightness levels with equal probability, a black and white TV transmitter transmits 25 frames per second. If the signal to noise ratio is 30dB, determine the minimum bandwidth required to transmit the video signal. Assume pixels to be statistically independent. | 3 | CO3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

7.	Consider a half width rectangular pulse, $\pi \left(\frac{t}{T_0} \right)$ Find and Sketch the Power spectrum Density of a. ON-OFF signalling b. Polar signalling	4	CO2
8.	A Random binary data sequence 110100101... is transmitted. Sketch the line coded version for a. Non-return to zero (L) b. Non-return to zero (I) c. Manchester d. Differential Manchester	4	CO2
9.	Answer ANY TWO of the following: i. The _____ line coding suppressed the DC Component. ii. The ARQ systems having the highest throughput efficiency is _____ and the lowest throughput efficiency is _____ iii. For a linear (7,4) block code, the ratio of parity bits to message bits is _____ and the code rate is _____ iv. Mutual Information between two discrete Random Variables X and Y is given by _____	2	CO4